

An end-to-end normal-crossing example: the model $N(x_0x_1, 1)$

Claude, with Daniel Murfet

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Abstract

Unit 218 follows Astra #24's top recommendation (programme H2-lite): after the hand-off, demonstrate applicability with one bounded example rather than enlarge the theorem catalogue. Observations $y_1, \dots, y_n$ are modelled as $N(x_0x_1, 1)$ with $x \in (-1, 1]^2$ and a continuous prior density ρ ; against the truth $N(0, 1)$ the likelihood ratio is *exactly* $\exp(-\frac{n}{2}(x_0x_1)^2 + \sqrt{n} Z_n x_0x_1)$ with $Z_n = n^{-1/2} \sum_i y_i$, i.e. the grammar paper's dressed normal-crossing kernel at $\beta = \frac{1}{2}$, $N = \sqrt{n}$, $h = 0$, $k = (1, 1)$, phase $2Z_n$ (`ncLikelihood_eq`). No resolution of singularities, no likelihood remainder and no empirical-process limit is assumed: the quadrant decomposition is Headline XIX's `symBox_phase_eq_sum` and the phase is a datum. The fluctuation variable is $\sqrt{n} x_0x_1$; its posterior moment generating function is a ratio of two model integrals with phases $2Z_n + 2\theta$ and $2Z_n$ (exponential tilt = phase shift), and the end-to-end theorem is

$$\mathbb{E}_{\text{post}}[e^{\theta \sqrt{n} x_0x_1}] \longrightarrow e^{z\theta + \theta^2/2} \quad \text{whenever } Z_n \rightarrow z$$

(`ncFluctMGF_tendsto`): the posterior law of $\sqrt{n} x_0x_1$ is asymptotically $N(Z_n, 1)$, centred at the empirical phase. Only $\rho(0) > 0$ is needed. Zero `sorry/axiom`; 9095 jobs.

1 The things formalised

```
theorem integral_quadKernel ( a ) : s, quadKernel a s = sqrt( / ) * exp ( a^2/4)
theorem phaseMoment_one_add_neg : phaseMoment 1 a + phaseMoment 1 (-
a) = sqrt( / ) e^{a^2/4}
theorem chart_const_phase_tendsto_moving : c → c , N → ∞
chartIntegral ... (N) (fun _ => c) / leadScale ... (N) → phaseCoeff ... (fun _ => c)
def ncLikelihood n y x := i, gaussianPDFReal (x 0 * x 1) 1 (y i)
def ncPhase n y := ( i, y i) / sqrt n
def ncKernel N z x := exp (-(N^2(x x)^2)/2 + N z x x); ncPosteriorMean; ncFluctMGF
theorem ncLikelihood_eq (hn : 0 < n) : ncLikelihood n y x =
( i, gaussianPDFReal 0 1 (y i)) * ncKernel (sqrt n) (ncPhase n y) x
theorem ncPosteriorMean_eq : posterior mean = ratio of kernel integrals
(the truth's likelihood factor cancels)
theorem nc_integral_tendsto :
({symBox 2} K_{N,z}) / leadScale ... → 2 (0) sqrt(2) e^{-{z^2/2}}
theorem ncFluctMGF_tendsto (h0 : 0 < 0) (hz : ncPhase n (y n) → z) :
ncFluctMGF n (y n) → exp (z + ^2/2)
```

2 Mathematics

Three ingredients on top of the existing block. (i) The *moving-phase* form of the leading-term theorem: with $c_n \rightarrow c_0$ and $N_n \rightarrow \infty$ the eventual Lipschitz dependence of the normalised chart integral on the phase (unit 209) and the fixed-phase limit (Headline XIII) give

$F_{N_n}(c_n) \rightarrow \mathcal{C}(c_0)$ — needed because the empirical phase moves with n . (ii) Completing the square: $J_1(a) + J_1(-a) = \int_{\mathbb{R}} e^{-\beta s^2 + \beta a s} ds = \sqrt{\pi/\beta} e^{\beta a^2/4}$, so the four quadrants ($\varepsilon_\sigma = \pm 1$ twice each) sum to $2\rho(0)\sqrt{2\pi} e^{z^2/2}$ at $\beta = \frac{1}{2}$, phase $2z$, and the ratio of the numerator (phase $2z + 2\theta$) to the denominator is $e^{((z+\theta)^2 - z^2)/2} = e^{z\theta + \theta^2/2}$. (iii) The exact likelihood identity, an elementary Gaussian computation. The example sits in the $m = 2$ logarithmic regime: the scale is $N^{-1} \log N$ and finite- n corrections are $O(1/\log n)$; a numerical check ($\rho = 1 + 0.3x_0 + 0.2x_1^2$, $z = 0.7$, $\theta = 0.5$, target 1.6080) gives 1.407, 1.498, 1.532, 1.559 at $n = 10^2, 10^4, 10^6, 10^8$ with $\text{gap} \times \log n \approx 1$ throughout, consistent with the theorem and with a $C/\log n$ correction.

3 Lean notes

- `integral_sub_right_eq_self` needs (`:= volume`) or the `IsAddRightInvariant` instance search is stuck on a metavariable.
- `Tendsto.div` produces a `Pi.div` of `lambdas`; before rewriting the pointwise identity, `simp only [Pi.div_apply]`. Pointwise goals from `integral_congr_ae` again need `beta_reduce` before `rw`.
- Sums over $\sigma : \text{Fin } 2 \rightarrow \text{Bool}$: `rw [← (piFinTwoEquiv fun _ => Bool).symm.sum_comp, Fintype.sum_prod_type]`; `simp only [Fintype.sum_bool, piFinTwoEquiv, Fin.cons_zero, Fin.cons_one, ...]` enumerates the four quadrants explicitly.
- The `ring` fallback message “Try this: `ring_nf`” is an info, not a warning, but the build log shows it; use `ring_nf` where `ring` only normalises.

4 Status

Headline XX (end-to-end normal-crossing example) added to the index. Next (unit 219): the Gaussian-data layer — for y_i i.i.d. $N(0, 1)$ the phase Z_n has variance one, so by Chebyshev and uniform convergence on compact phase sets the difference $\mathbb{E}_{\text{post}}[e^{\theta \sqrt{n} x_0 x_1}] - e^{Z_n \theta + \theta^2/2}$ tends to zero in probability.